

PesTwin



The biology-informed Digital Twin transforming Integrated Pest Management.

18th September 2025

Climate-Sensitive Vector Dynamics

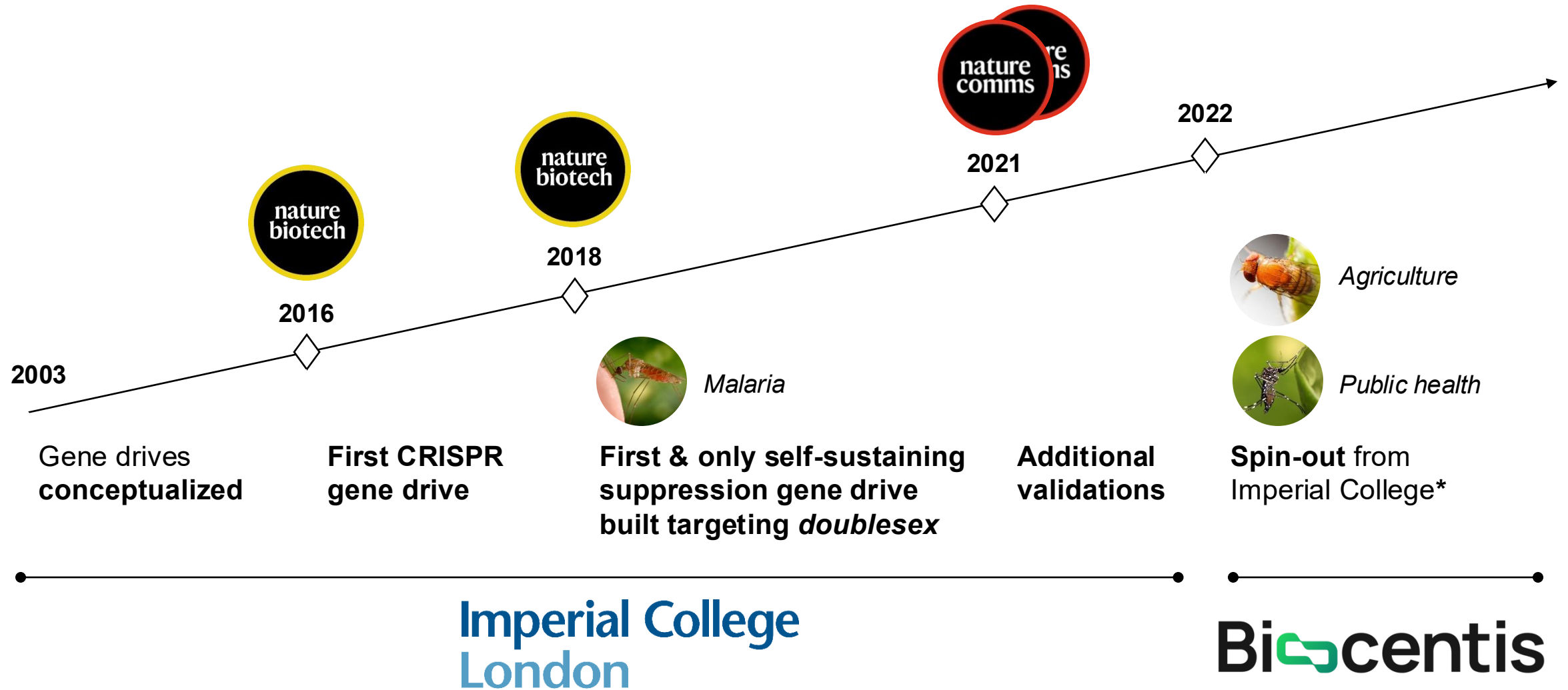
Modelling Workshop, Nicosia

Andrea De Antoni

Biocentis s.r.l.

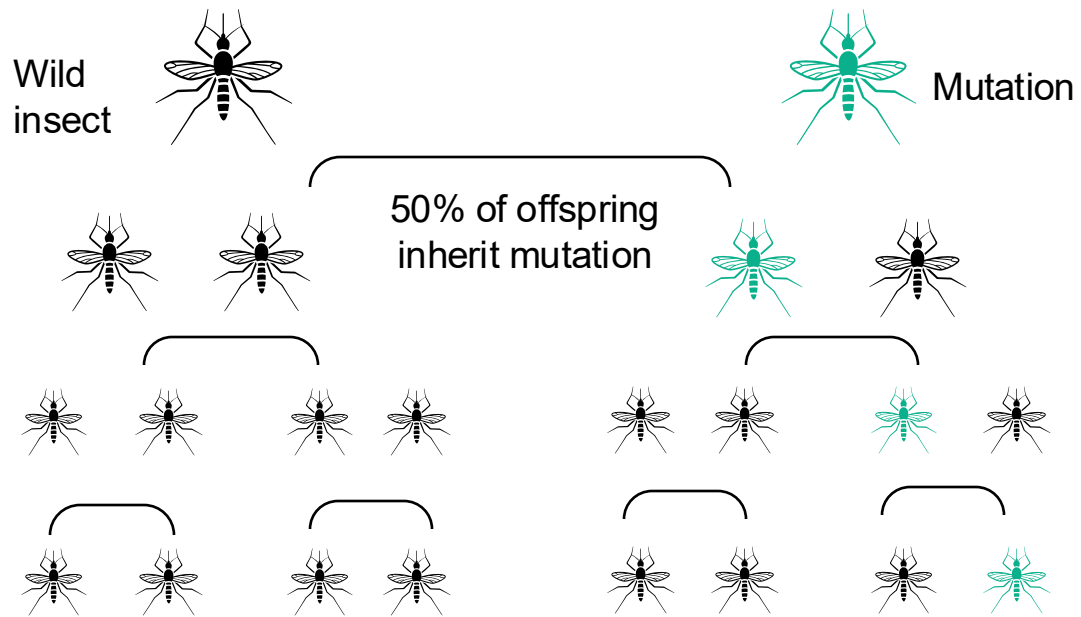
Industrial Innovation PhD Program, **Università di Trento**¹

Biocentis is a startup working on innovative genetic pest control technologies

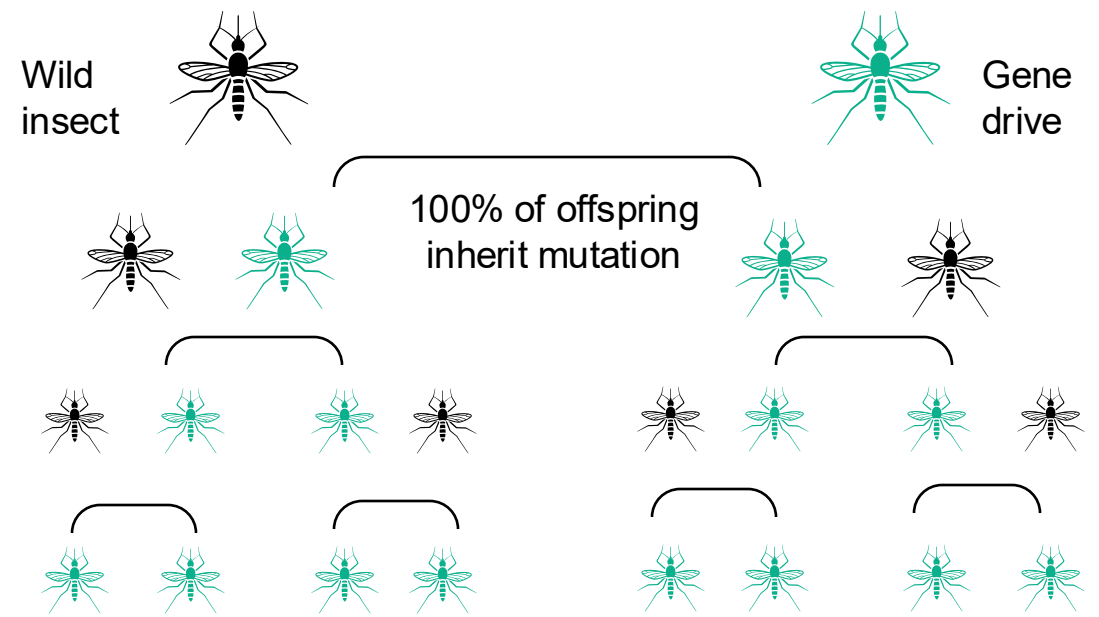


Gene-drives are a powerful tool that can propagate desirable traits in a target population

Mendelian inheritance



Gene drive



Current R&D focus is on self-limiting technologies (in time and space), not gene-drives

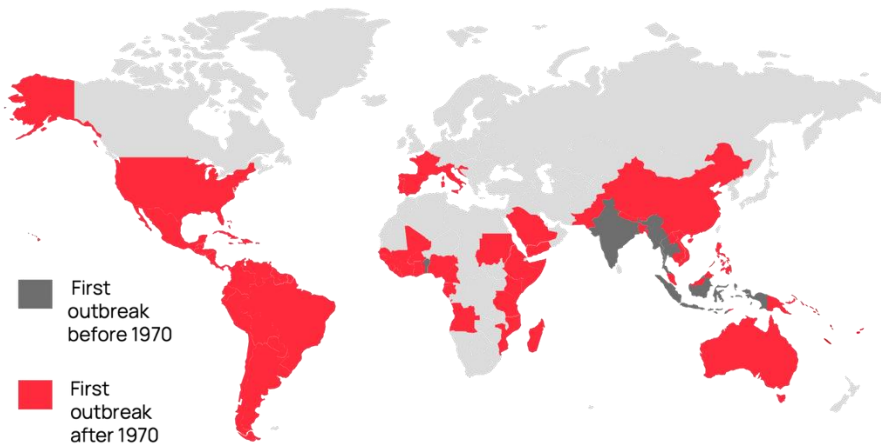
R&D focus is on two highly invasive species

Public Health

Aedes aegypti

(1970-present)

expanded globally, driving dengue transmission



>4B *Global population at risk of infection*

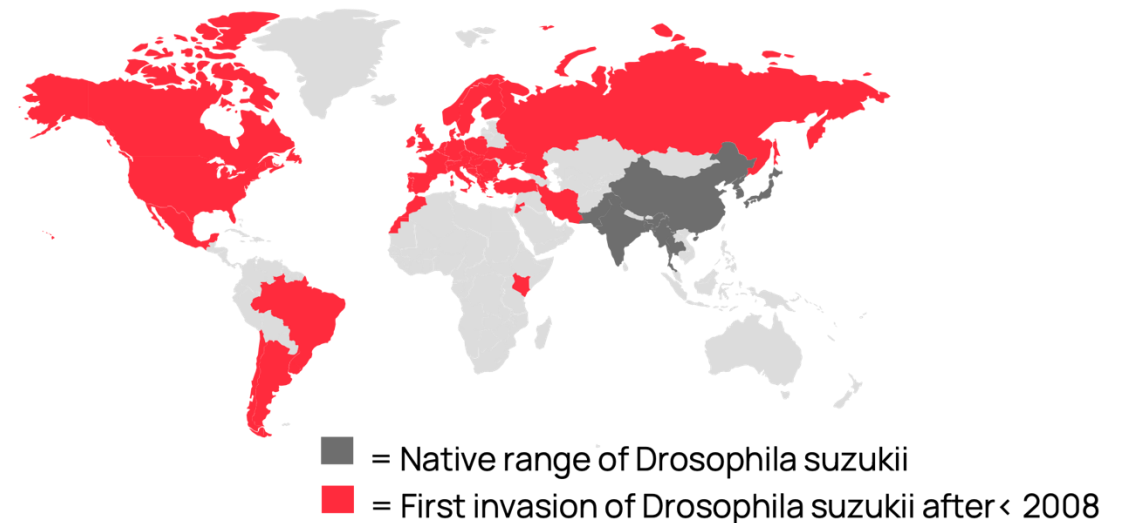
\$40B *Estimated yearly economic impact*

Agriculture

Drosophila suzukii

(2008-present)

became a global threat to the berry and cherry industries



~\$4B *Estimated yearly crop losses worldwide*

What is the role of modelling in a company such as Biocentis?

Lab Experiment Design



Biologist

Use modelling to evaluate population dynamic outcomes of new genetic constructs at the early stage of design

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Support Business Development



Business Developer

Use modelling to explain Biocentis technology core concepts to investors or commercial partners

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Regulator

Use modelling to integrate Lab data dossier to demonstrate the bio-safety of new genetic control technology

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Field Release Strategy Design



IPM Manager

Use modelling to design pest suppression field interventions, reduce costs, and manage risks

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developing **PesTwin**

a modelling & simulation framework focusing on insect pest control strategies

From the upcoming support to the
design of experiments in the Lab..

..to a visionary digital core
of an ***IPM Decision Support System***

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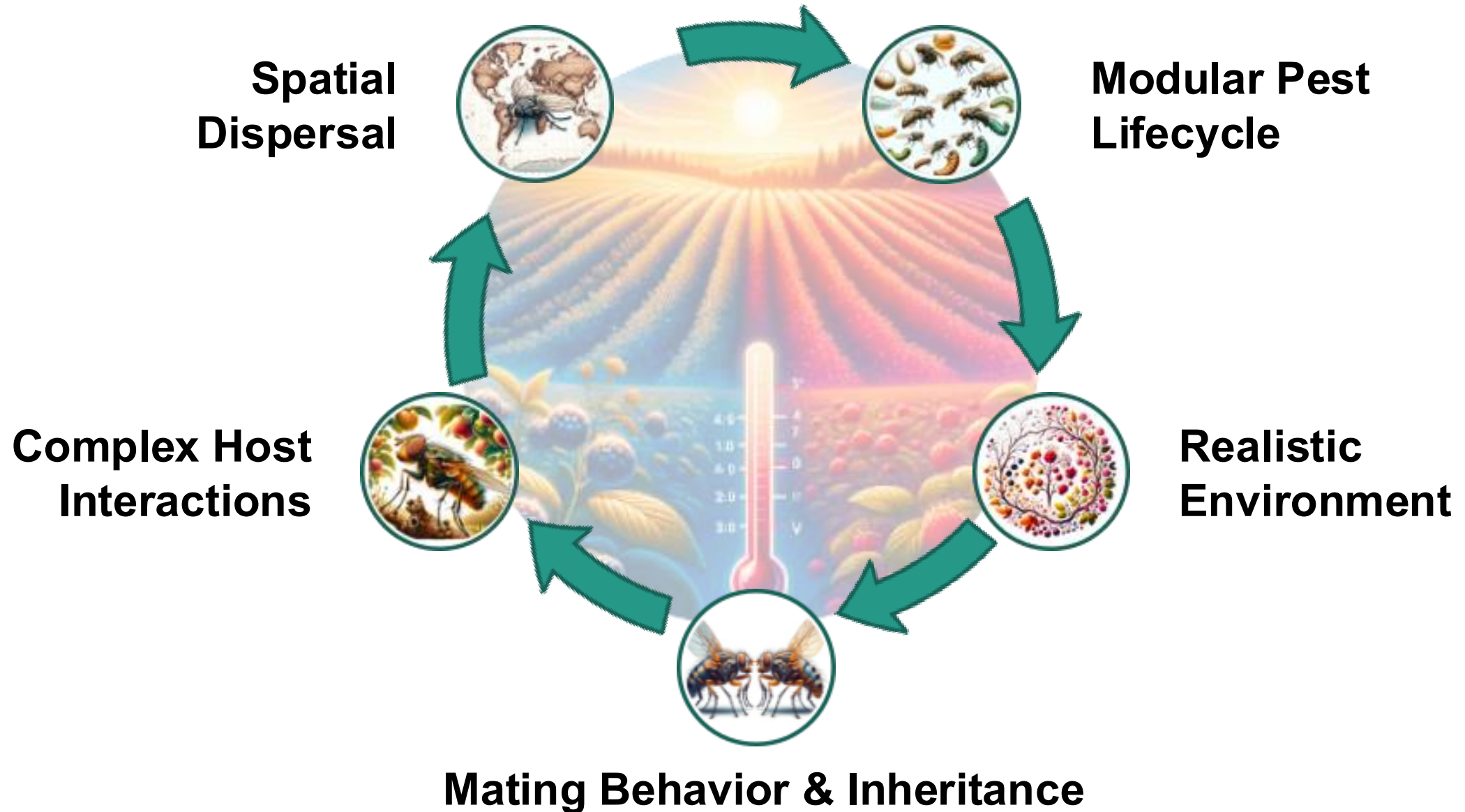
Field Release Strategy Design



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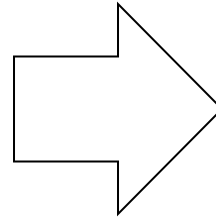
Functional requirements driving the design of PesTwin simulator software



Agent Based Modelling and Simulation (ABMS) to handle the complexity of realistic ecology

Functional Requirements:

- Modular Pest Lifecycle
- Realistic Environment
- Mating Behavior & Inheritance
- Complex Host Interactions
- Spatial Dispersal



Agent Based Modelling & Simulations

based on **FLAME GPU**, an accelerated simulator for domain independent complex systems simulations.

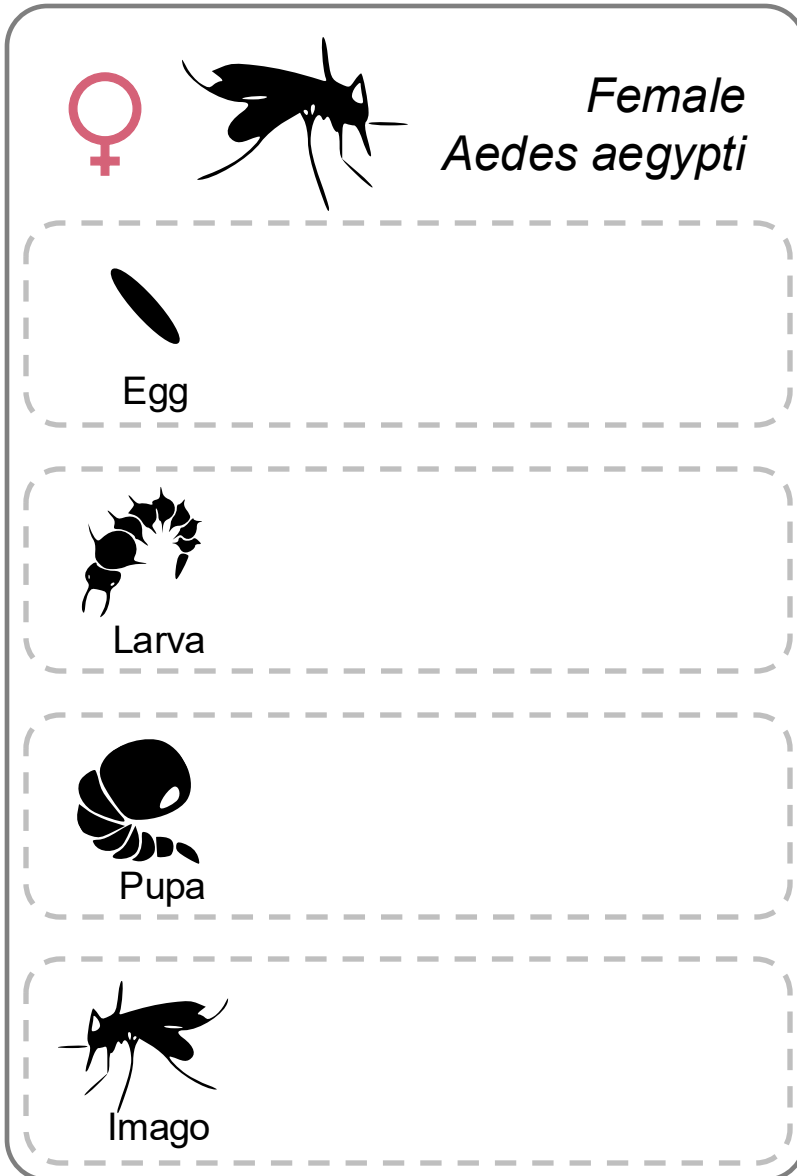


Modularity and Testability at the core of the architecture design



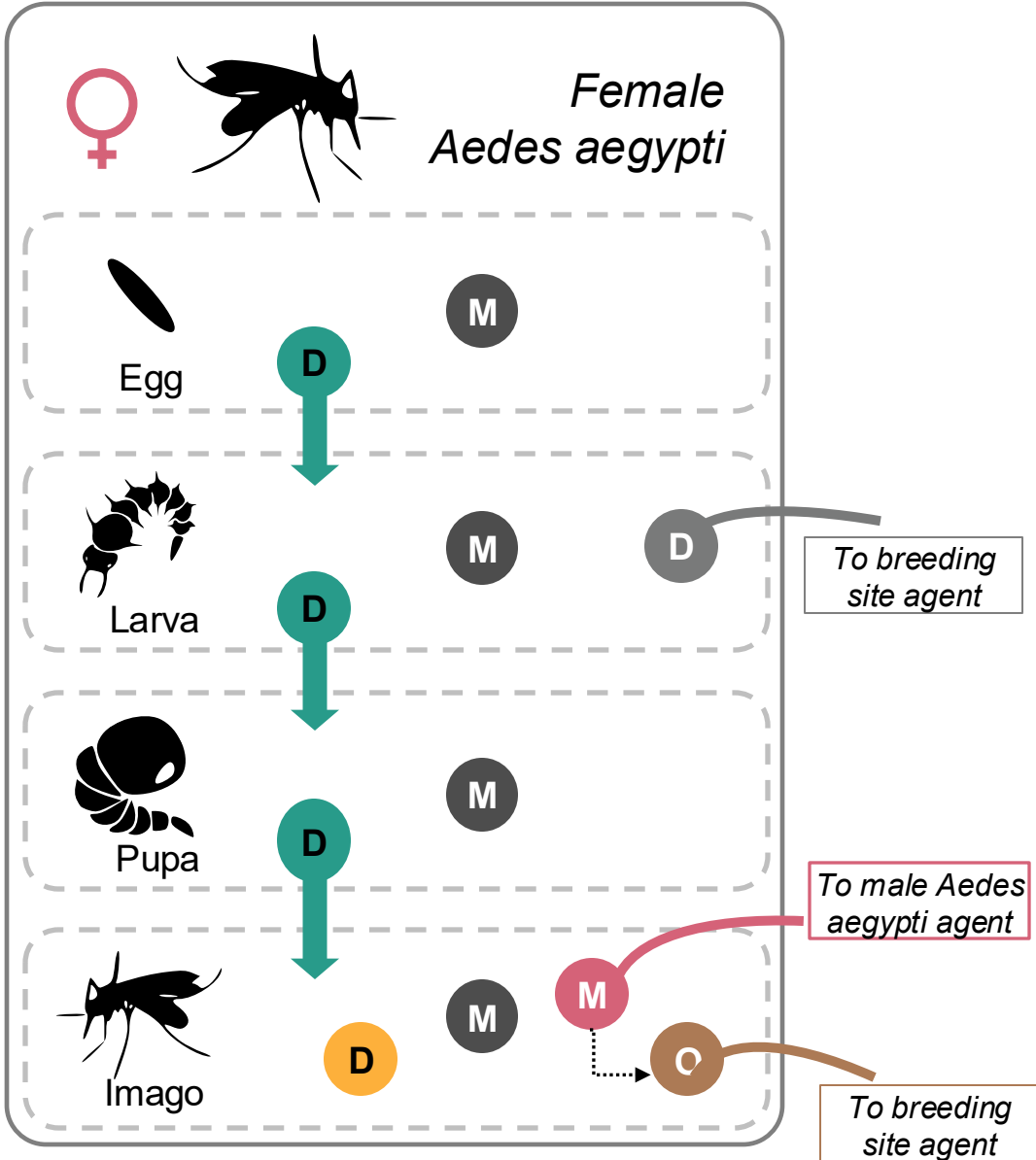
- **Agents** are autonomous computational entities interacting with themselves and their environment

Modularity and Testability at the core of the architecture design



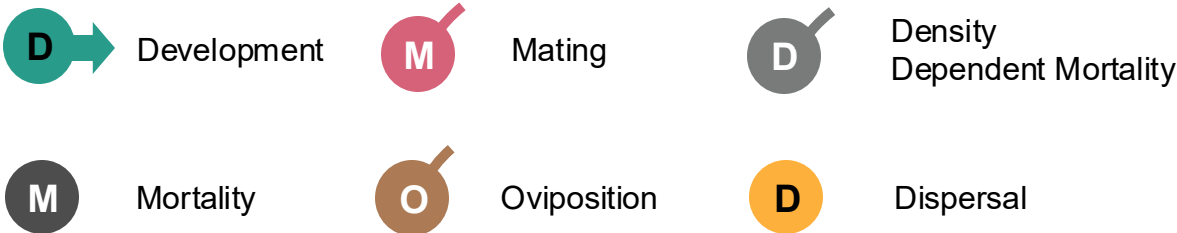
- **Agents** are autonomous computational entities interacting with themselves and their environment
- **States** compose the lifecycle of each agent, with transitions from one to another

Modularity and Testability at the core of the architecture design



- **Agents** are autonomous computational entities interacting with themselves and their environment
- **States** compose the lifecycle of each agent, with transitions from one to another
- **Life Processes** are modular and configurable behavioral rules which can be attached to agent states, defining how agents act and interact

Life Processes Library



Legend



Relational Process

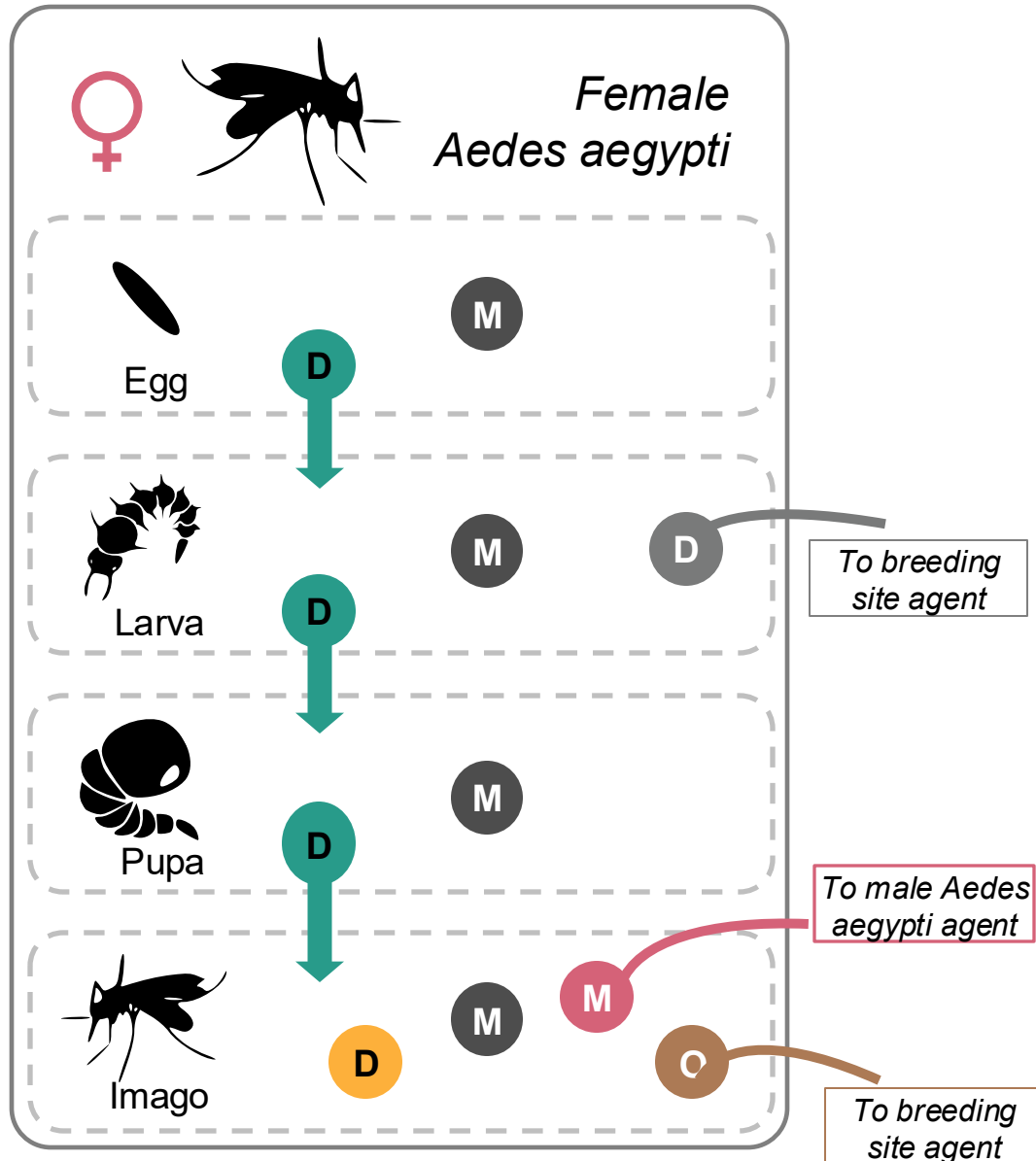


State-Change Process



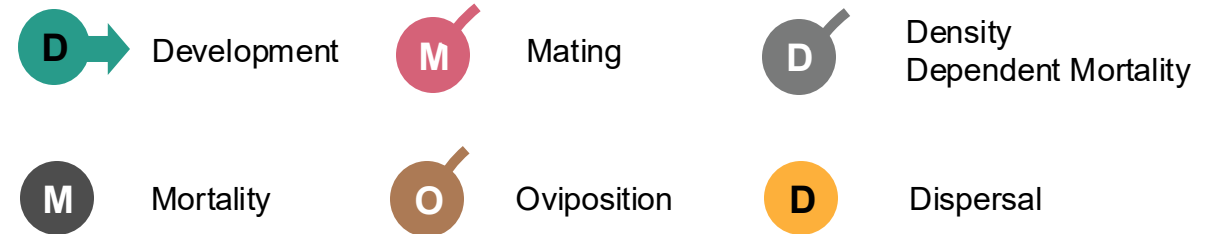
Behavioral Change

Modularity and Testability at the core of the architecture design



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Life Processes Library

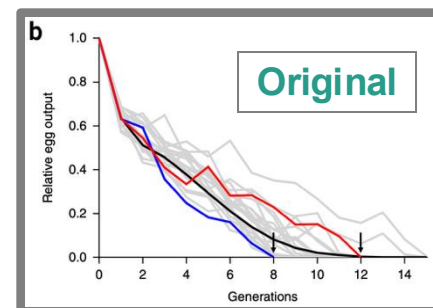
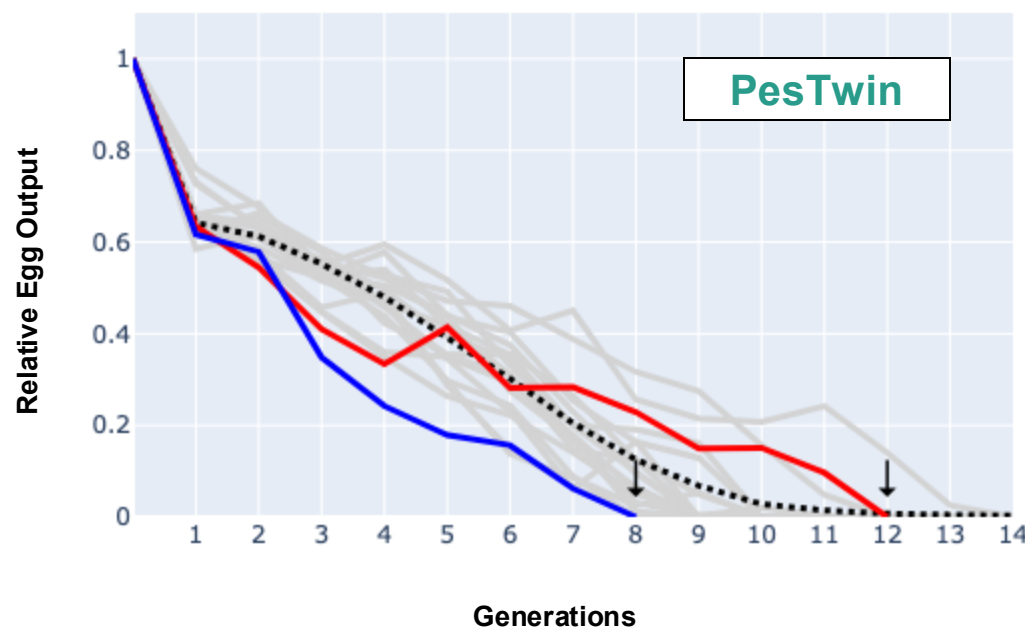


- **Process Functions** define different functional implementations of the behavior of each life process

Example: Erlang vs Exponential Development process implementation

Validation 1: Population suppression in small cages

Modelling Results



Egg Output decreases as the Gene Drive construct spreads into the population. Experimental cage populations get suppressed in 8 and 12 generations, respectively.

Use Case



Insect:
Anopheles gambiae

Paper Ref:
(Kyrou et al., 2018)

Scenario:
Pest Suppression in Small Cages

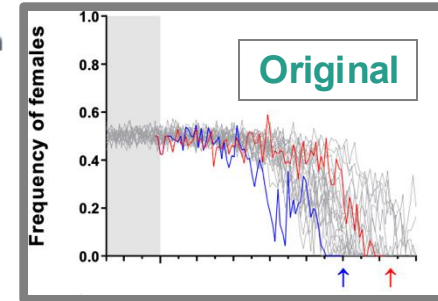
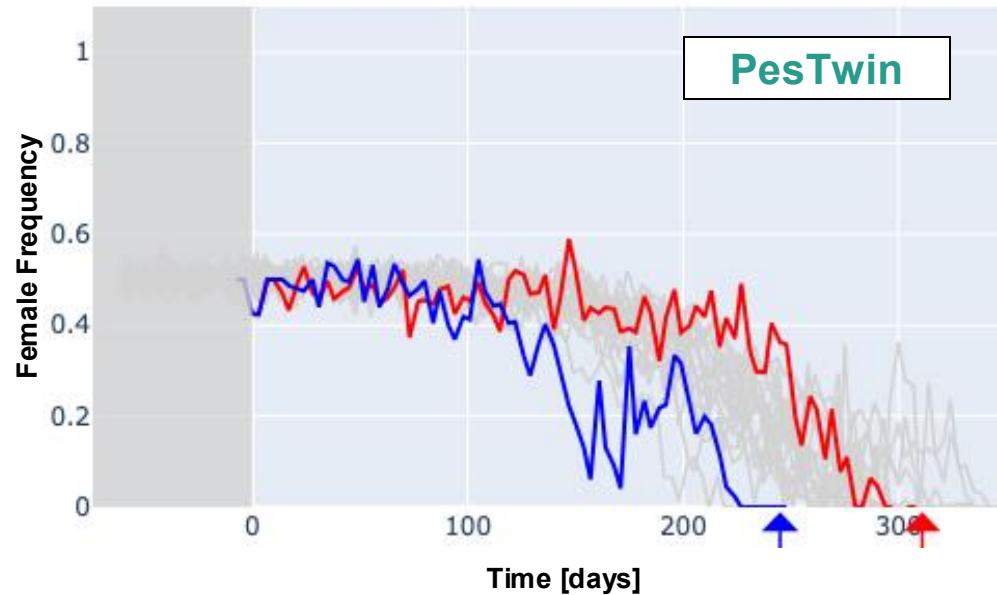
Control Technology:
DSX Gene Drive

Modelling Feature

- **Genetic Modifiers:** the success rate of each life process, such as oviposition or mating, depends on the agent's genotype. Genotype inheritance is modeled via an inheritance cube.

Validation 2: Population suppression in large cages

Modelling Results



Wildtype Female Frequency decreases as the Gene Drive construct spreads into the population. Experimental cage populations are suppressed between days 250 and 300 after transgene release.

Use Case



Insect:
Anopheles gambiae

Paper Ref:
(Hammond et al., 2021)

Scenario:
Pest Suppression in Large Cages

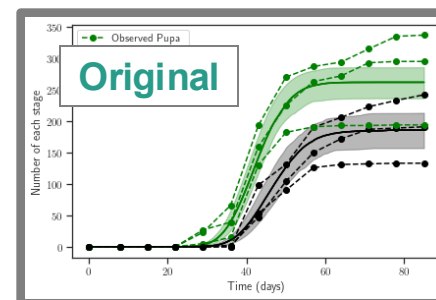
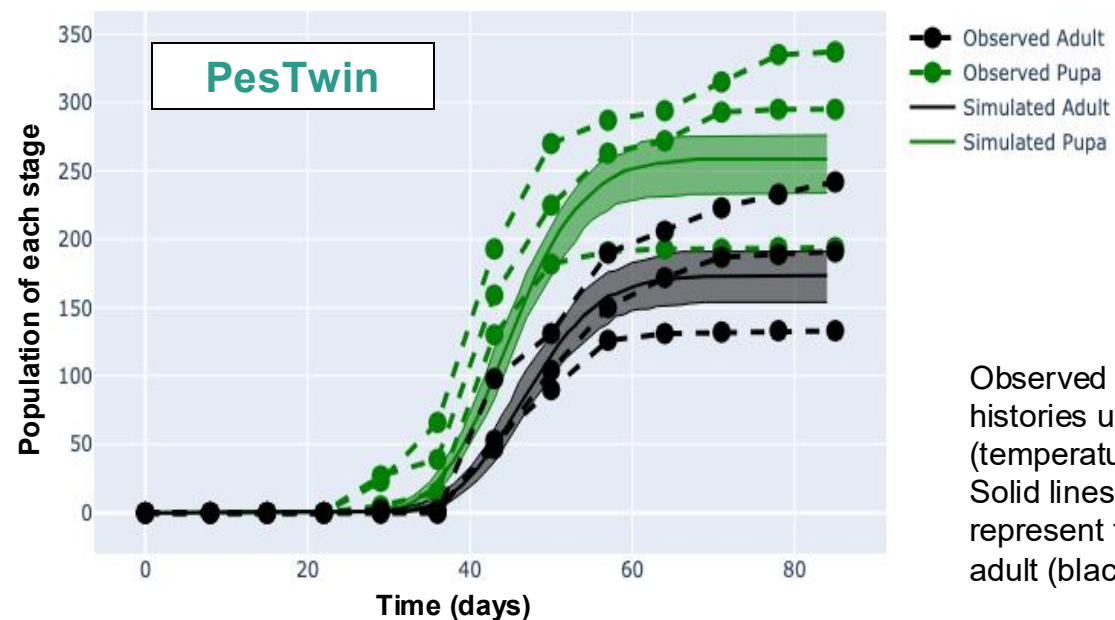
Control Technology:
DSX Gene Drive

Modelling Feature

- **Epigenetic Modifiers:** the agent's parents' genotypes contribute to defining the success rate of the agent's life processes, such as female fertility.

Validation 3: Population growth in climatic chambers

Modelling Results



Observed and simulated mosquitoes' life histories under constant temperature conditions (temperature = 15 °C). Solid lines represent the median, shaded areas represent the 90% range of pupa (green) and adult (black) population.

Use Case



Insect:
Culex quinquefasciatus

Paper Ref:
(Erguler et al., 2022)

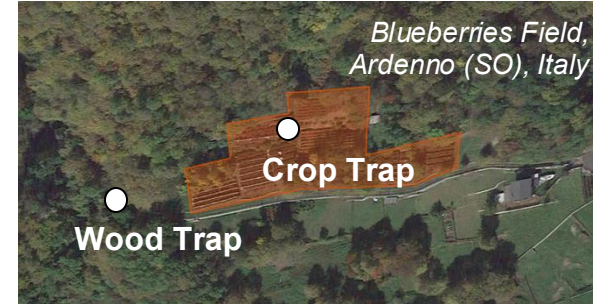
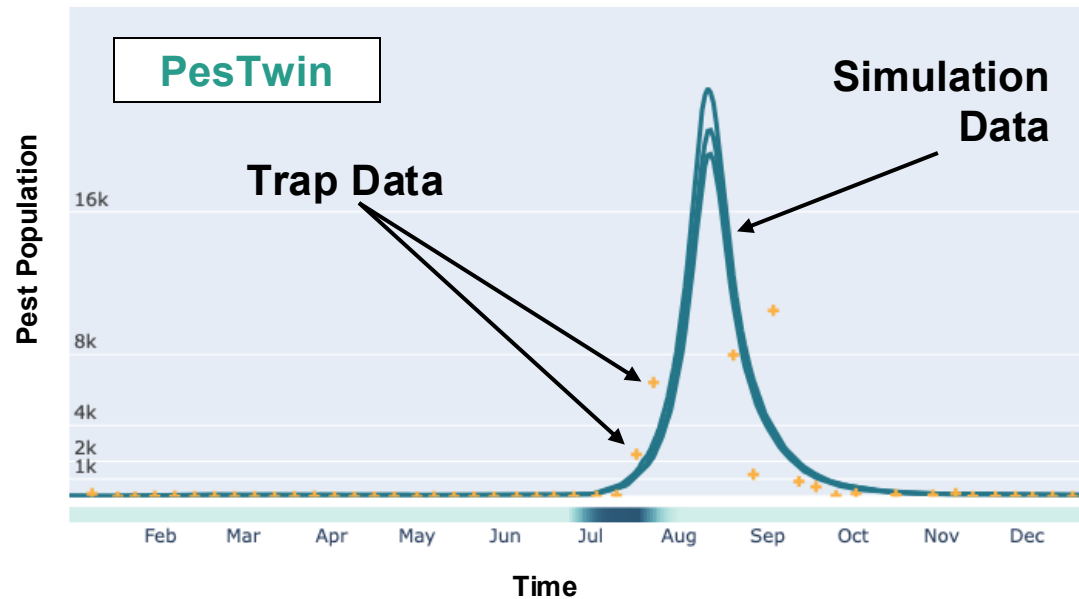
Scenario:
**Pest Population Growth
in Climatic Chambers**

Modelling Feature

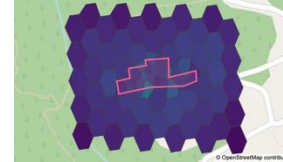
- **Temperature-dependent lifecycle:** insect development and mortality as a renewal process with variable rates depending on environmental conditions.

Validation 4: Pest monitoring in open field

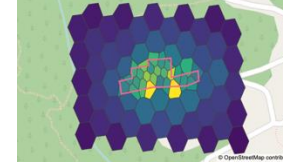
Modelling Results



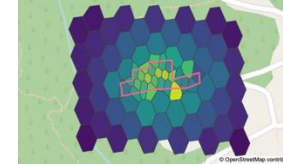
July 20th



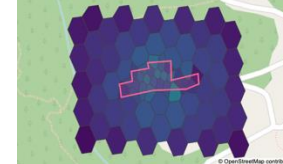
July 30th



Aug 9th



Aug 19th



Use Case



Insect:
Drosophila suzukii

Paper Ref:
(in preparation)

Scenario:
**Pest Monitoring in an
Open Field**

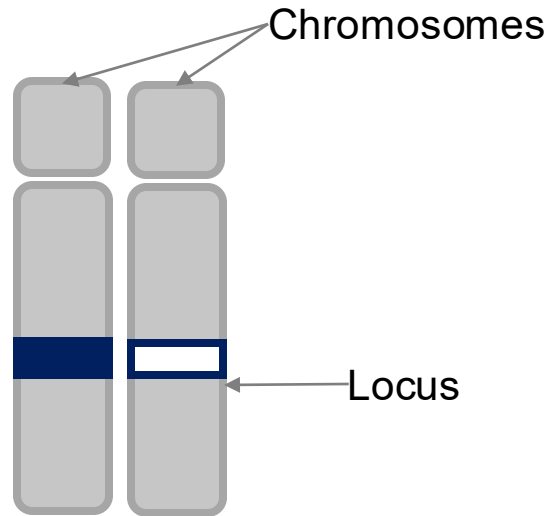
Modelling Feature

- **Dispersal:** agents can move over the edges of a dispersal graph, interacting with the other agents who share the same location.

Agents can be equipped with genetic information, transmitted to offspring via genetic inheritance rules

M

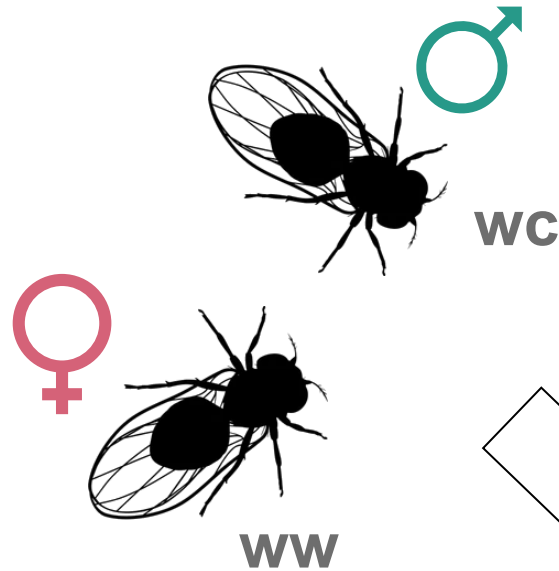
Diploid Genome



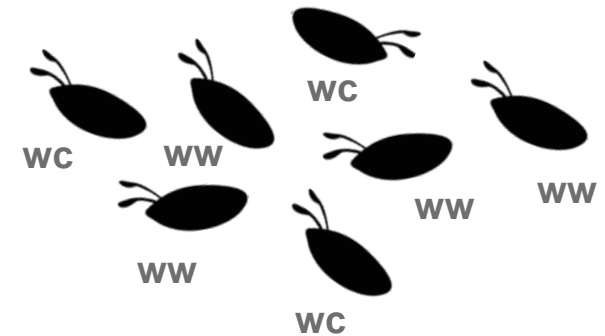
Alleles = {w, c}

Genotypes = {ww, wc, cw, cc}

Phenotypes = {ww, wc, cc}



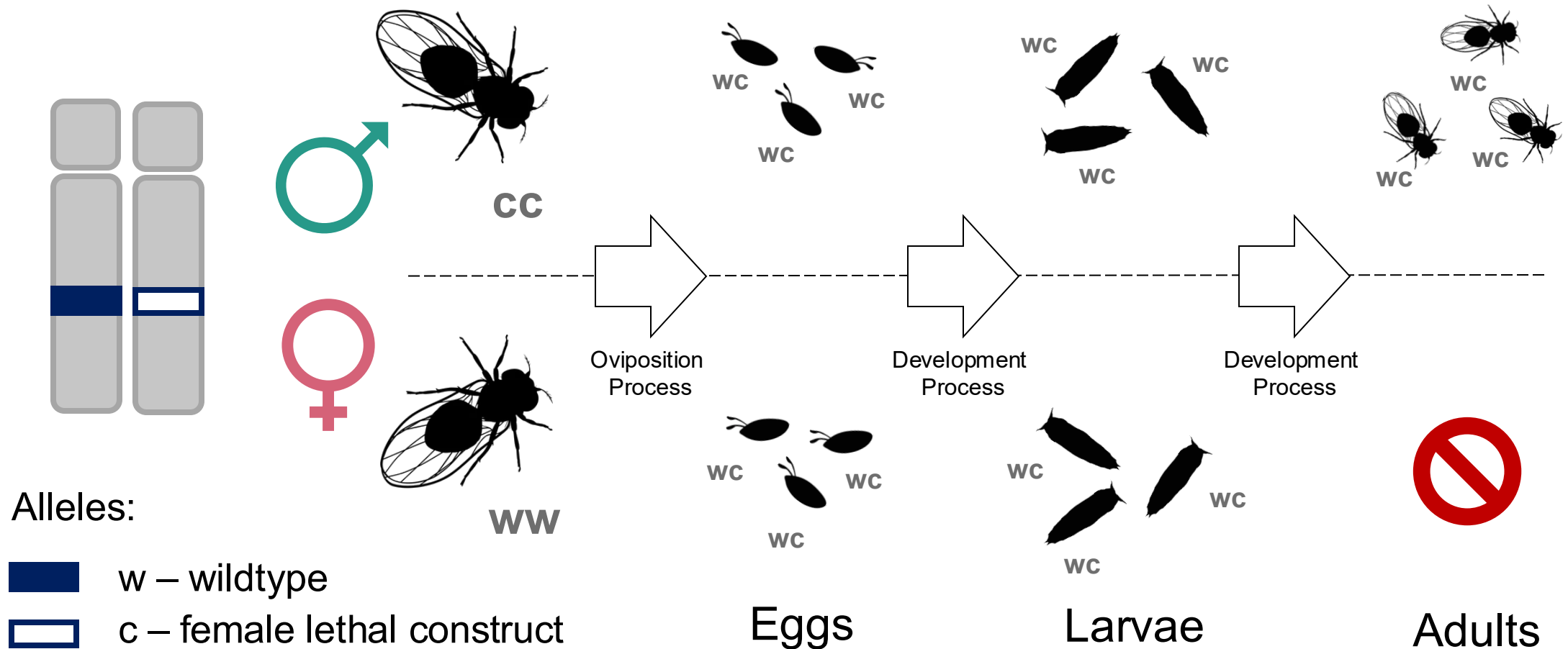
WW			
	WW	WC	CC
WW	1	0.5	0
WC	0.5	0.25	0
CC	0	0	0



$$P(\text{Offspring} = ww \mid \text{Mother} = ww \ \&\& \ \text{Father} = wc) = 0.5$$

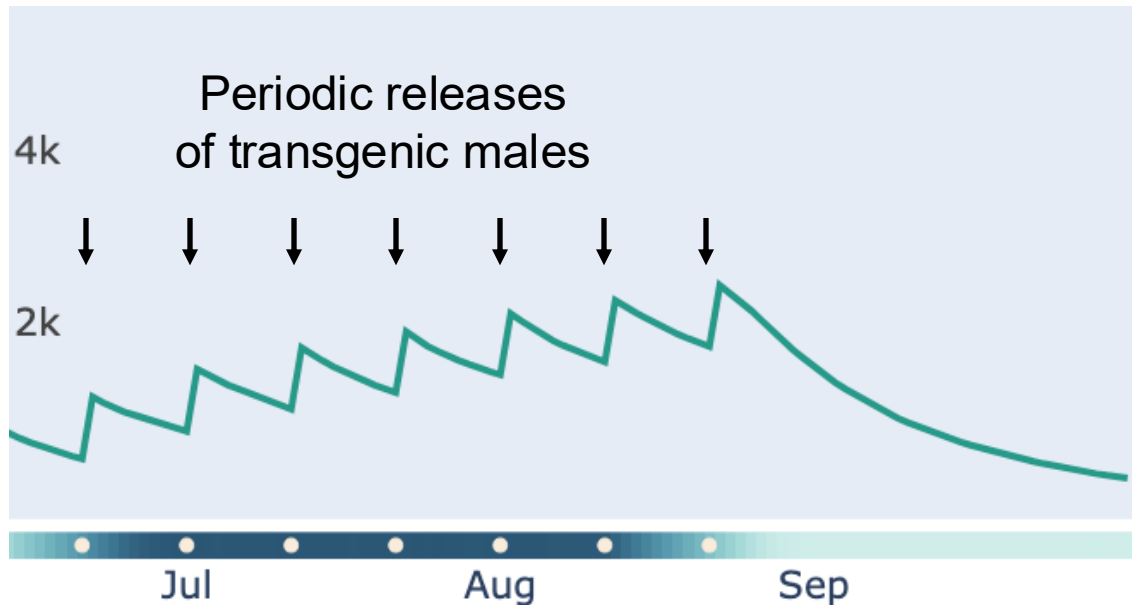
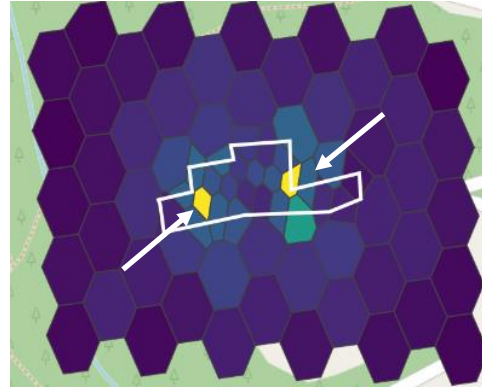
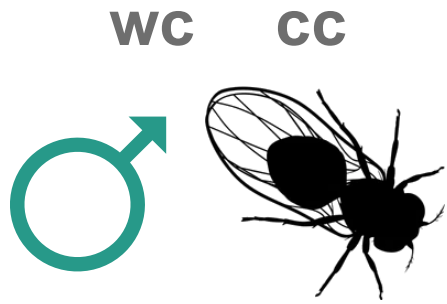
Agents can be equipped with genetic information, influencing agent phenotypes as multipliers applied to agent process success rate

fsRIDL: female-specific Release of Insects carrying Dominant Lethal



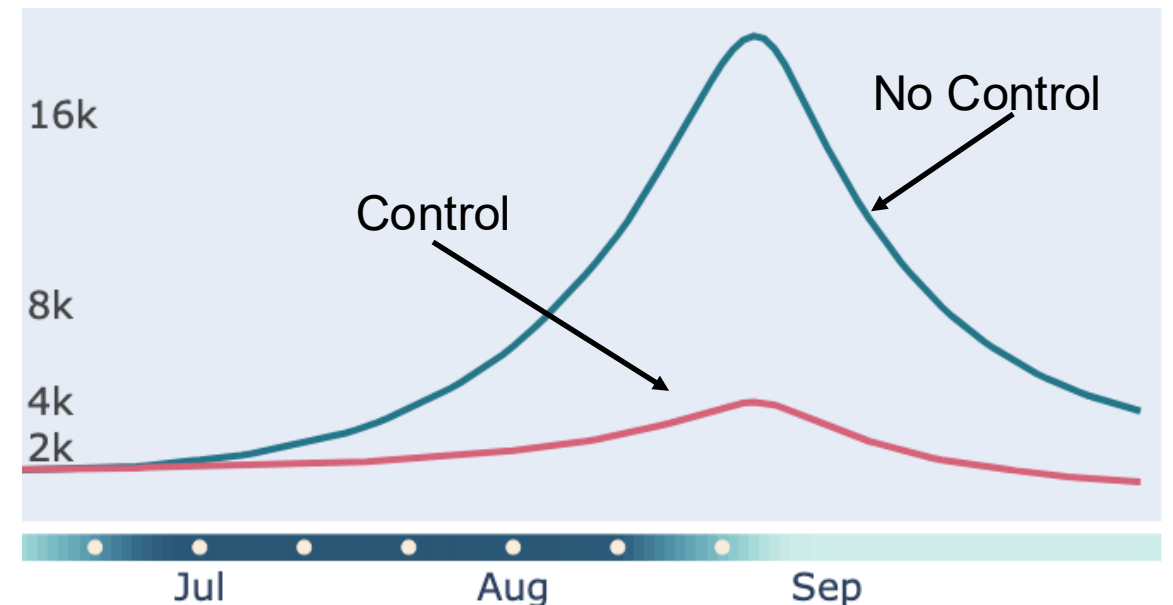
Simulations can be used to plan, evaluate and optimize (any) IPM control strategy tailored to a specific field

Transgenic Males



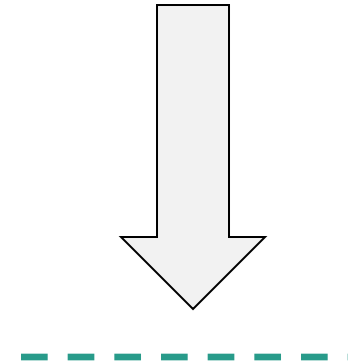
Wildtype Females

Evaluate the efficacy of a control strategy



Pestwin Technology Readiness Level: 4 to 5 TRL

TRL	Description
1	Basic Principles Observed
2	Technology Concept Formulated
3	Experimental Proof of Concept
4	Technology Validated in Lab
5	Technology Validated in Relevant Environment
6	Technology Demonstrated in Relevant Environment
7	System Prototype Demonstration in Operational Environment
8	System Completed and Qualified
9	Actual System Proven in Operational Environment

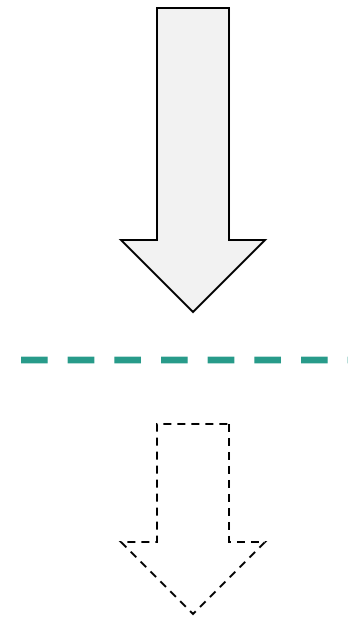


NODES Project
(01-2024 – 06-2025)
Conceptualization, architecture design
and first PoC development

(06-2025 – present)
Validation against literature data,
preparing for application in the Biocentis
Lab

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Conceptualization, architecture design
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Looking for **scientific / research partnerships** to:

- advance technology maturation
- refine existing use cases
- explore new opportunities

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(Not-exhaustive) list of topics of interest:

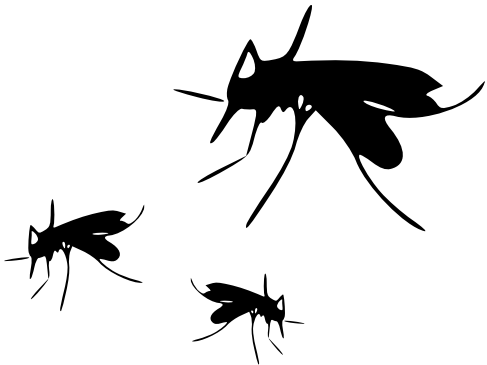
- Improve modelling details:
 - Polyandry, diapause, photoperiod, etc.
- Other control strategies:
 - Chemicals, parasitoids, IIT, etc.
- Improve model accuracy & calibration:
 - Semi-field conditions, dispersal, etc.
- Modelling (and data) standards for interoperability

Acknowledgments



Matteo Rucco,
Previous Data Science Team Leader at Biocentis

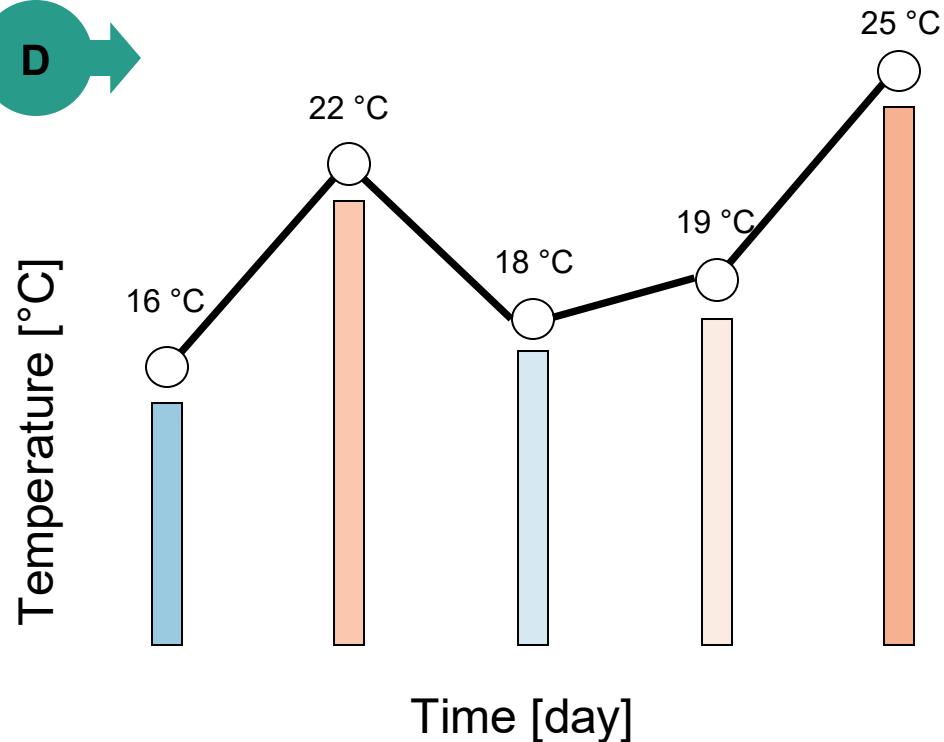
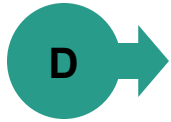
Question Time!



Bibliography

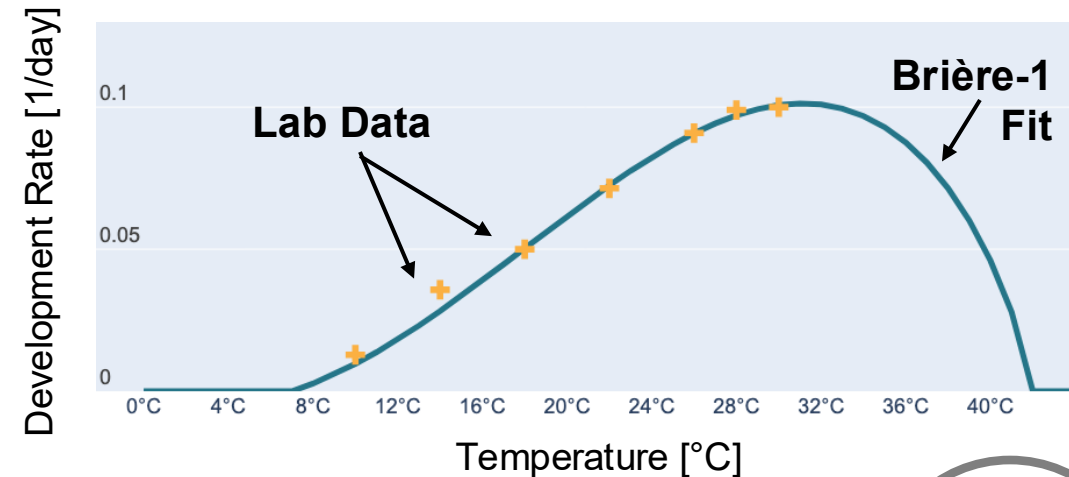
- Erguler, K., Mendel, J., Petrić, D. V., Petrić, M., Kavran, M., Demirok, M. C., Gunay, F., Georgiades, P., Alten, B., & Lelieveld, J. (2022). A dynamically structured matrix population model for insect life histories observed under variable environmental conditions. *Scientific Reports*, 12(1). <https://doi.org/10.1038/s41598-022-15806-2>
- Hammond, A., Pollegioni, P., Persampieri, T., North, A., Minuz, R., Trusso, A., Bucci, A., Kyrou, K., Morianou, I., Simoni, A., Nolan, T., Müller, R., & Crisanti, A. (2021). Gene-drive suppression of mosquito populations in large cages as a bridge between lab and field. *Nature Communications*, 12(1). <https://doi.org/10.1038/s41467-021-24790-6>
- Kyrou, K., Hammond, A. M., Galizi, R., Kranjc, N., Burt, A., Beaghton, A. K., Nolan, T., & Crisanti, A. (2018). A CRISPR–Cas9 gene drive targeting doublesex causes complete population suppression in caged *Anopheles gambiae* mosquitoes. *Nature Biotechnology*, 36(11), 1062–1071. <https://doi.org/10.1038/nbt.4245>

Pest development is modelled as a renewal process with a dynamic temperature-dependent probability

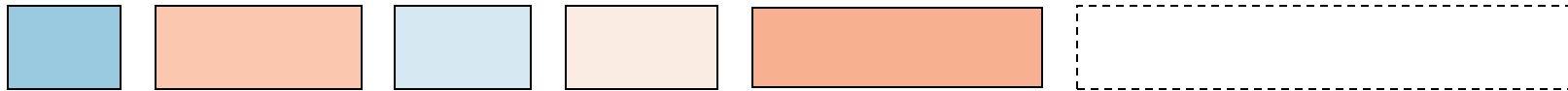


Erlang-distributed process, the mean linked to daily temperature via **Brière-1 function** model.

$$\mu = Br(T)^{-1} = \left(\alpha T(T - T_0) \sqrt{T_1 - T} \right)^{-1}$$



Larva

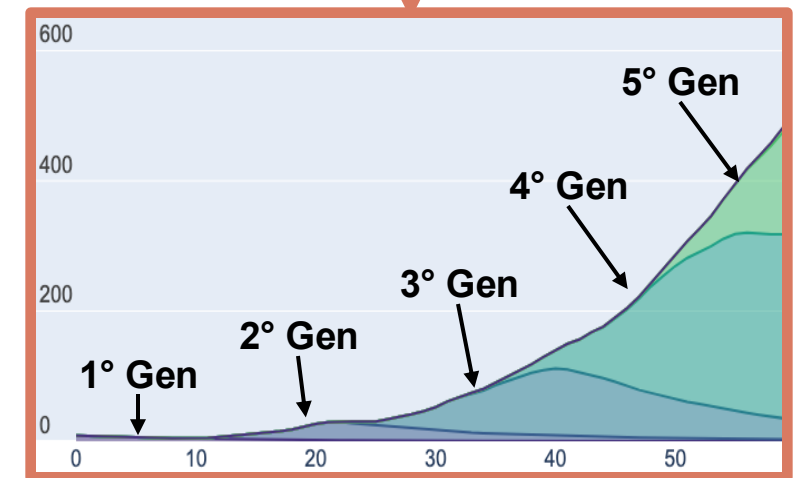
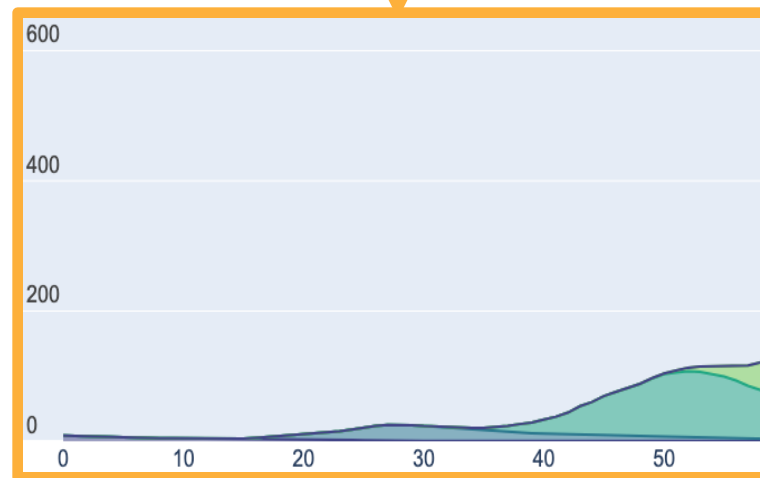
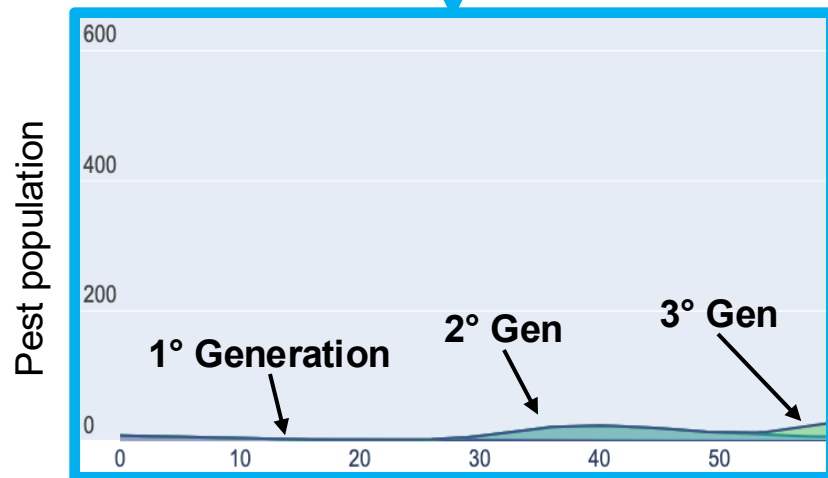
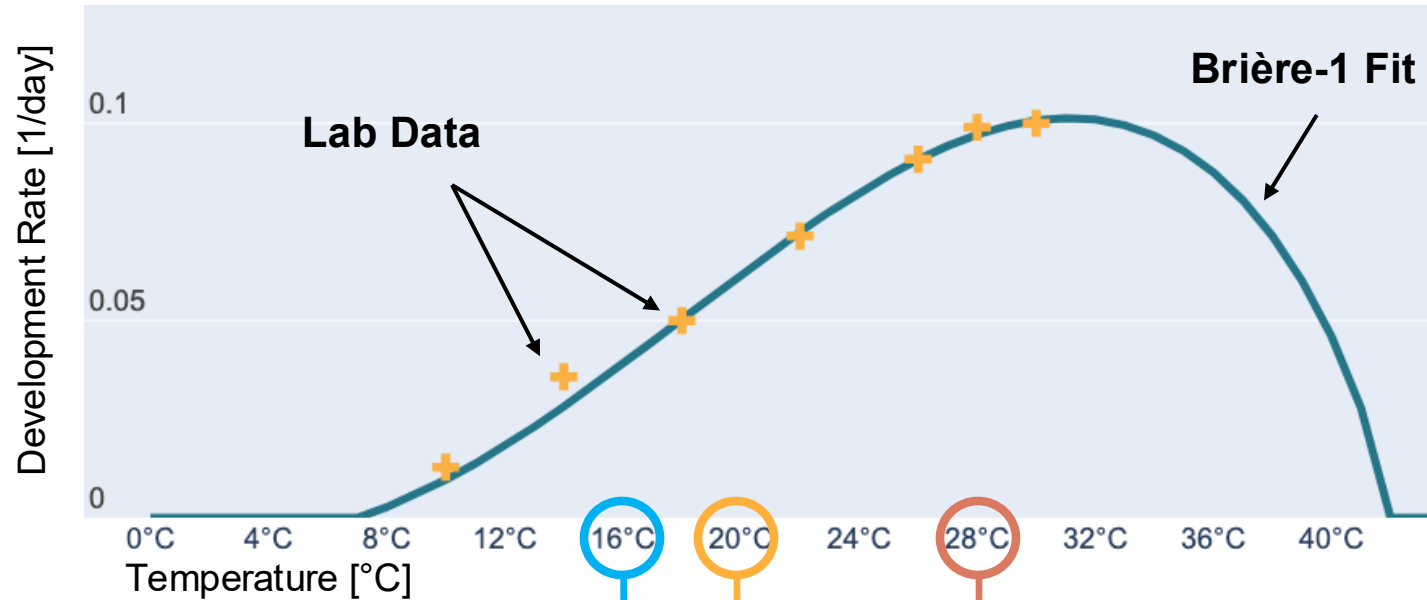
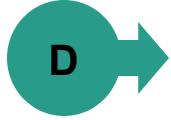


Temperature-dependent stage development algorithm
adapted from [Kamil Erguler & al. \(2022\)](#).



Adult

Warmer temperatures leads to faster pest development, determining an exponential growth of the population during the ripening season



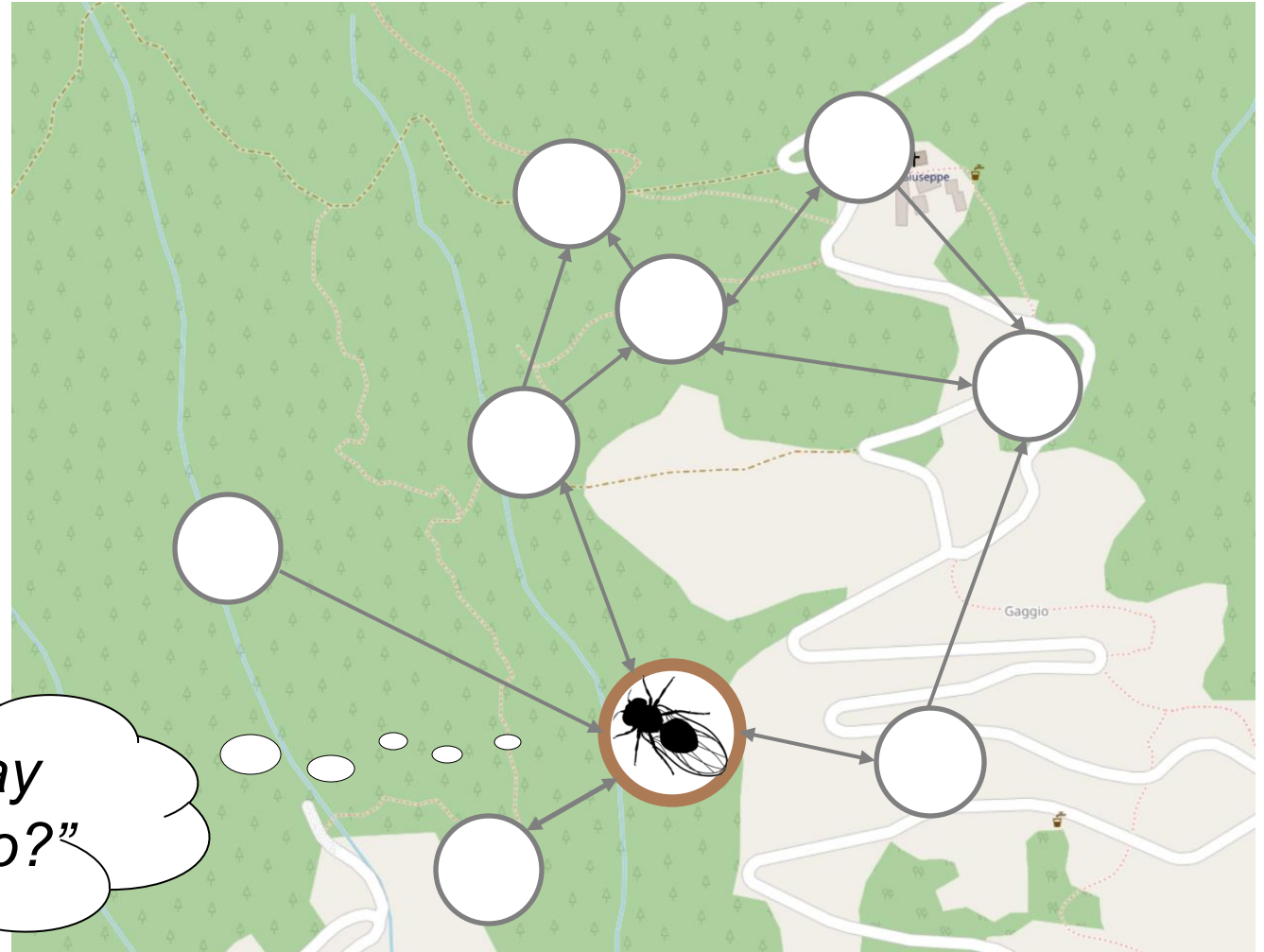
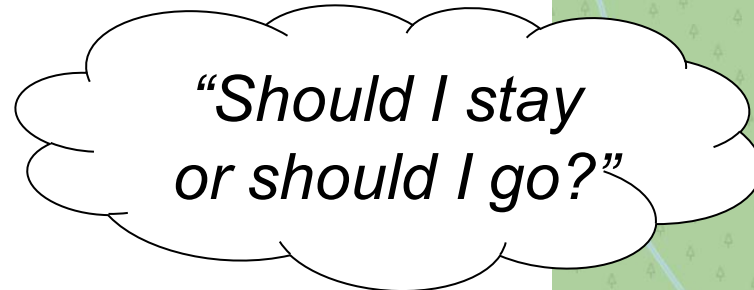
Time [day]

Space is represented as a direct graph: agent-to-agent interactions happen between agents sharing the same location

D

Agent dispersal is a two-steps process:

1. **Exit** step, modelling the probability of the agent to leave its current site,

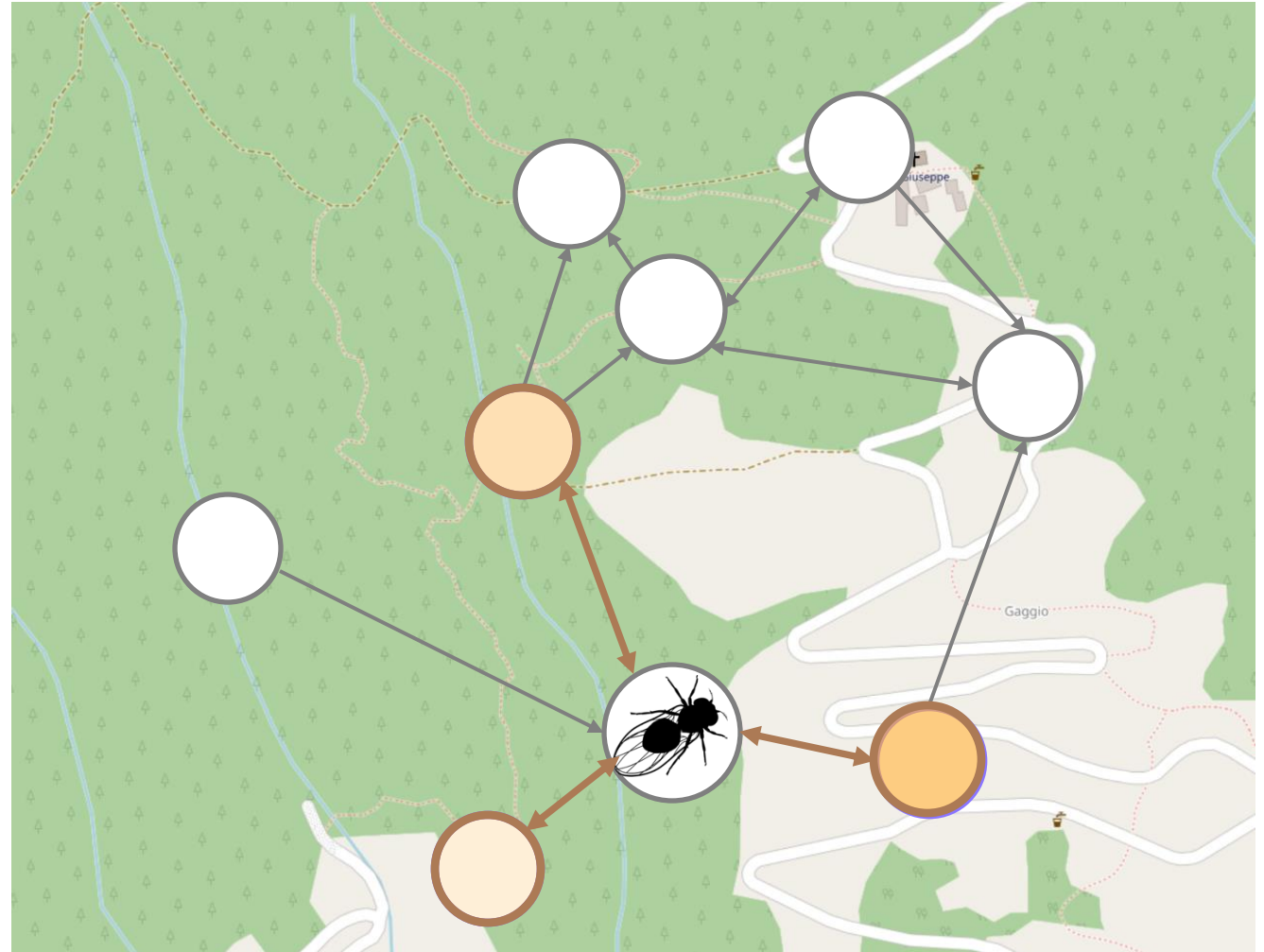


Space is represented as a direct graph: agent-to-agent interactions happen between agents sharing the same location

D

Agent dispersal is a two-steps process:

1. **Exit** step, modelling the probability of the agent to leave its current site,
2. **Move** step, modelling the probability for the agent to choose a specific graph edge exiting the current site.

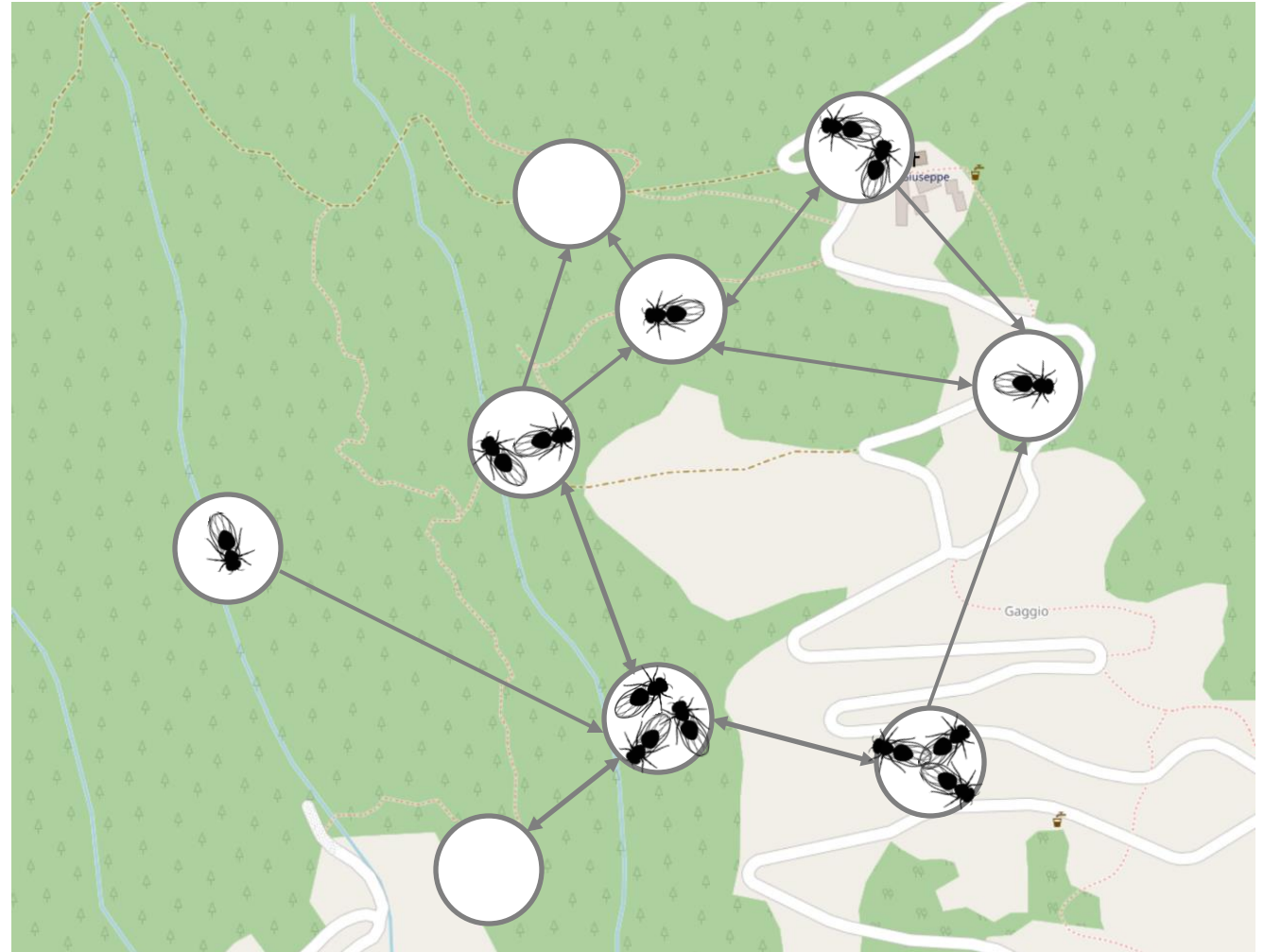


Space is represented as a direct graph: agent-to-agent interactions happen between agents sharing the same location

D

Multiple functional forms:

- inhomogeneous landscapes dispersal,
- resource-seeking movements,
- short/long distance dispersal,
- wind assisted dispersal



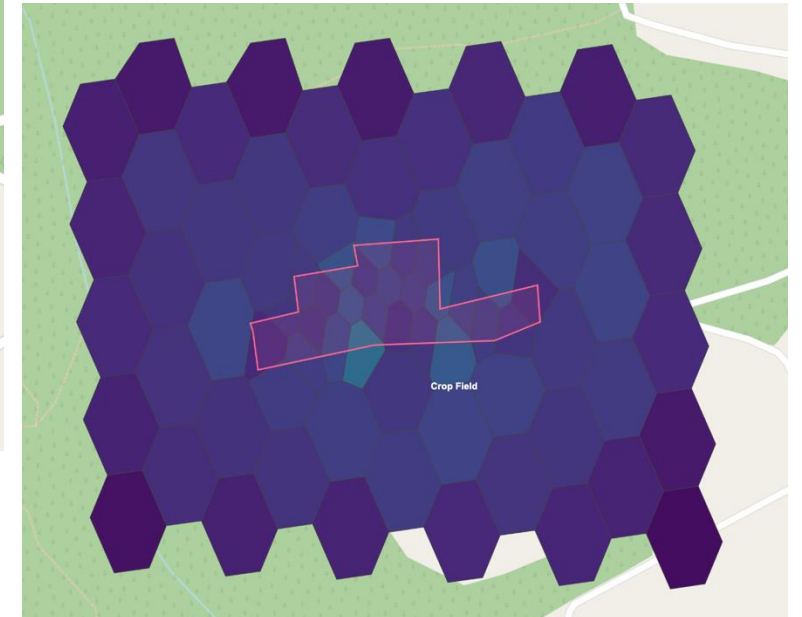
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Field GIS data..

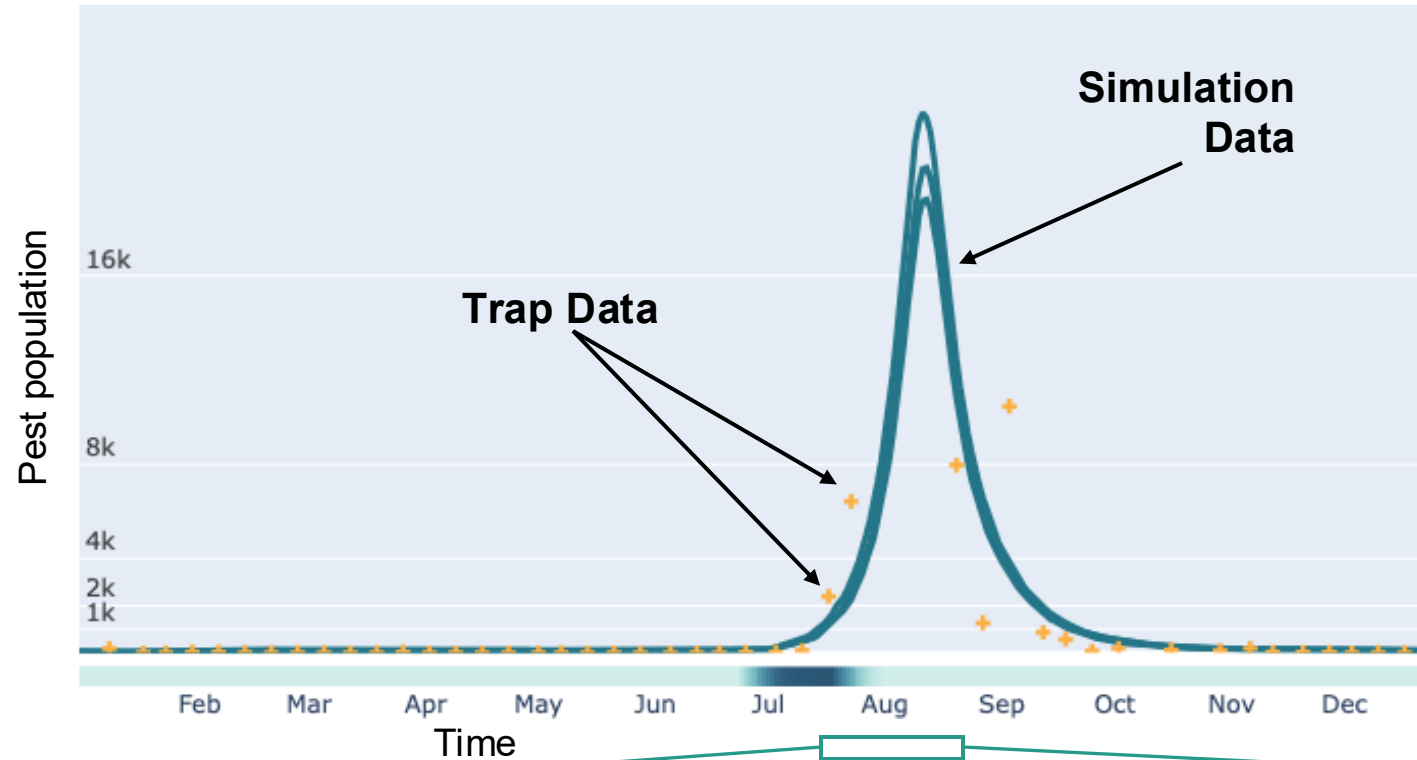


.. agents dispersal graph..



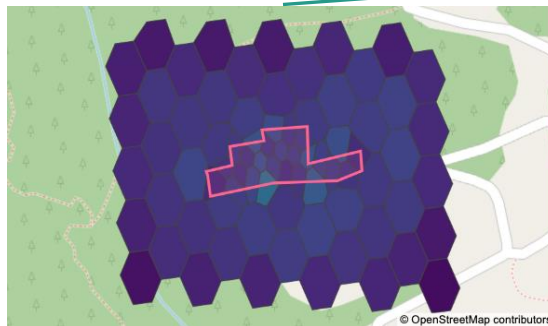
..spatially informed simulation.

Coupling trapping data, pest biology and temperature profile, we get a spatiotemporal simulation of the infestation

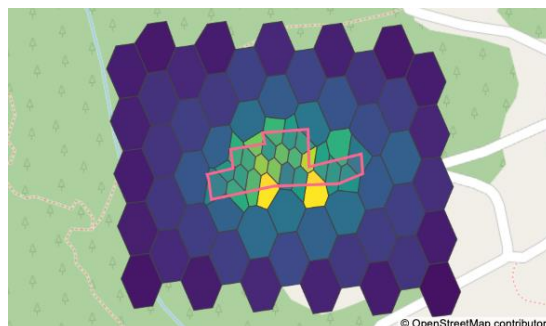


Simulator's qualitative validation on field data.

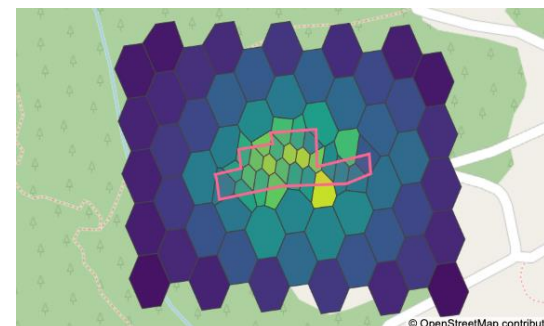
Host Fruit
In Ripening Stage



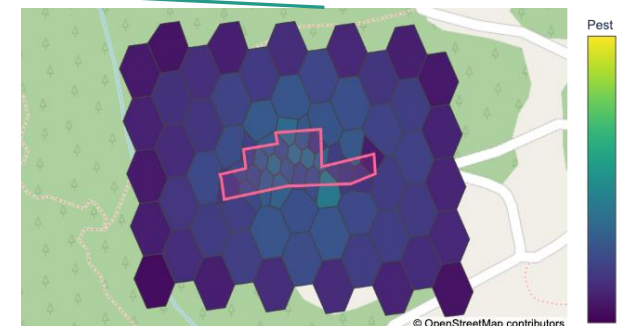
July 20th



July 30th



August 9th



August 19th